

AI Assistant for Reliable Knowledge: Supporting Student Learning through Lecture Notes with Retrieval-Augmented Generation

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CHAPTER 1

INTRODUCTION

Many students still struggle to grasp complex academic content, which can lead to suboptimal learning outcomes across various educational contexts. In higher education, the sheer quantity and complexity of lecture notes often overwhelm students without immediate contextual assistance, making it difficult for them to process and retain information effectively, which can hinder students' understanding and academic performance (Binkley, 2023). Though lecture notes are designed to provide students with detailed information, it can often hinder rather than support effective learning. Cognitive Load Theory (CLT) explains that learners have a limited working memory capacity, and when instructional materials are too lengthy or poorly structured, they impose unnecessary intrinsic and extraneous cognitive load. This overload prevents students from processing information efficiently and transferring knowledge into long-term memory. In the context of online learning, where students rely heavily on digital lecture materials, extended notes without proper segmentation or instructional design may lead to confusion, fatigue, and disengagement. Studies have shown that learning content quality strongly influences both e-learning quality and student satisfaction, highlighting the importance of managing cognitive load in instructional design. Thus, instead of aiding comprehension, extended lecture notes may unintentionally reduce learning outcomes by overwhelming students and diminishing their ability to retain essential concepts. (Abeysekera et al., 2024)

Artificial intelligence has emerged as a transformative force. With the rise of large language models, the landscape of artificial intelligence has significantly transformed, with its applications spreading across various domains as these models are characterized by their ability to generate human-like text. This has revolutionized tasks such as natural language understanding, translation, summarization, and question answering (QA) that were previously constrained by the limitations of traditional rule-based systems and smaller task-specific models. Advances in LLMs have led to an increase in practical use, offering real-time support to help students clarify their doubts and enhance their learning experience (Roumeliotis & Tselikas, 2023). However, while LLMs have significant benefits, they present problems and limitations. A notable drawback is the risk of the model hallucinating, where the AI generates responses that are incorrect, nonsensical, or unfaithful to the provided source content (Ji et al., 2023). Additionally, because LLMs primarily relies on learned data, it may be challenging to update or extend their knowledge base in real-time (Oche et al., 2025).

A solution that can be utilized is Retrieval-Augmented Generation (RAG). A hybrid approach that combines the generative power of LLMs with the ability to retrieve real-time information from external sources, such as databases or the

internet. This hybrid approach grounds that the responses generated by the AI to be accurate and up to date to the information retrieved, thus reducing the likelihood of hallucinations and improving the overall reliability of the system (Lewis et al., 2020). RAG holds promise in educational settings, where it is critical for student to have accurate, contextually relevant information to support their learning (Swacha & Gracel, 2025). However, while there are numerous studies on enhancing LLMs through RAG, but research into its practical application in the education sector, particularly in helping students understand and learn from their university lecture notes, remains underexplored.

This opens an opportunity to explore the topic. We introduce Qubo, a retrieval-augmented question answering system designed to support student learning through accurate and contextually grounded responses. The system relies on RAG to integrate a knowledge base wherein it seeks relevant information that helps it retain factual consistency to ensure students still retain knowledge from the sources. This study covers the design and development of Qubo, we aim to investigate its effectiveness in providing accuracy and reliability in providing learning support and assess whether this AI-driven system can be a reliable alternative form of learning method for students.

1.1 BACKGROUND OF THE STUDY

Use of AI or artificial intelligence is becoming more and more prevalent in schools to assist with such things as personal learning automated grading and even paper handling. Research indicates that AI has enormous potential for increasing student productivity and accessibility but also poses fears over the effect it will have on education. Problems such as less human contact, privacy concerns, algorithm bias, and unequal access to technology need to be confronted. To reap the most of AI without succumbing to its harms, Al-Zahrani (2024) suggests a well-balanced strategy that couple's innovation with ethics, teacher training, and openness.

Students have increasingly started relying on AI-powered chatbots. Large language models (LLMs), such as OpenAI's ChatGPT, provide instant explanations, feedback, and answers to questions. Many students find this helpful because it aids them in asking about topics, they have a hard time understanding. Research also shows that these chatbots can boost engagement and confidence, especially in online courses where teacher interaction is limited [8]. For example, in a study on an R programming course, students reported that the chatbot helped them solve complex problems not covered in lectures, and many even requested similar tools for other classes [4]. During the first year of the COVID-19 pandemic, researchers analyzed how U.S. high school students reflected on their educational experiences. A lot of students reported difficulties with time management, friendships, and mental health. Most also said their academic performance, motivation, social life, self-control, and job preparedness were negatively affected. Still, students noted some benefits, such as becoming more self-reliant, improving their use of digital technologies, and developing a greater appreciation for their professors. The study further showed that while the class of 2023 and students from less advantaged families faced greater struggles, effective educational support and opportunities for in-person learning helped reduce these challenges [18].

Another study looked at how the mental health of freshmen affects their academic performance. It found that students struggling with mental health issues often received lower grades. On the other hand, those who adjusted well to college life and believed in their own academic abilities were better able to overcome challenges. In short, good mental health, adaptability, and confidence in learning work together to help students succeed [17]. However, LLMs also come with challenges. One of the biggest problems is what researchers call "hallucinations," where the model generates answers that sound convincing but are wrong or misleading. This creates serious issues when used in sensitive areas such as education, healthcare, or privacy. Ji et al. (2023) argue that better evaluation methods and stronger safeguards are needed to make these systems more reliable [2].

A promising solution to this problem is Retrieval-Augmented Generation (RAG). Instead of relying only on what the model has learned during training, RAG combines LLMs with external sources of accurate information, grounding responses in real data. Early studies show that RAG-based chatbots are more reliable but also effective for education, since they provide course-specific and trustworthy support [4]. This makes RAG an important step forward in using AI responsibly in education, offering students better learning experiences while addressing the weaknesses of traditional LLMs, such as hallucinations, outdated knowledge, and transparent reasoning when implemented properly. By connecting LLMs to outside information sources, Retrieval-Augmented

Generation (RAG) can address these problems and potentially improve accuracy and validity.[19].

This study explores how AI technologies help in modern learning challenges through the utilization of RAG in the development of Qubo, a chatbot specialized for source-grounded student learning. This study will be valuable not only to students seeking alternative, reliable learning support but also to educators exploring supplementary approaches to facilitate student learning. The insights derived from this study can be used to widen the knowledge on how to effectively use AI for reliable learning. Moreover, the research can help shape the future development of AI-assisted learning tools, offering educators a deeper understanding of how these systems can be used to foster independent learning and improve engagement among students. The study strives to facilitate a much better understanding of how supplementary AI tools can be integrated into the education sector to help enhance student learning. This study is especially relevant in the context of the Philippines, where education is highly valued, yet access to quality educational resources remains a challenge in many regions. By examining the potential of Qubo and similar AI tools, this study aims to highlight how AI-powered learning platforms can serve as an accessible and scalable solution to enhance student engagement and academic performance, particularly in regions where traditional learning methods may fall short.

1.2 STATEMENT OF THE PROBLEM

The increasing complexity and volume of information in modern education present significant challenges for students trying to effectively learn and retain key concepts from lectures. While lecture notes are a key resource, they are often dense, incomplete, or lack explanations that are easy to understand, limiting students' ability to grasp complex topics and slowing down their learning. Specifically, the problems are as follows:

1. Difficulty in Understanding Lecture Notes

Students frequently struggle to learn from lecture notes alone because the content can be fragmented and lacks sufficient context, making it difficult to connect concepts and fully grasp complex topics.

2. Limited Access to Immediate Clarification

The traditional learning process is often reliant on students' manual review of notes and does not provide immediate support from the Instructor. This leaves students with unanswered questions that can hinder their overall understanding.

3. Overload and Fragmentation of Learning Resources

The sheer volume and scattered nature of academic content can overwhelm students, leading to a weak understanding of core ideas and continuing knowledge gaps.

4. Inefficient and slow learning

The challenge in navigating dense and incomplete notes slows down the learning process, making it difficult for students to absorb information and retain key concepts.

5. Overwhelming Teacher Workload

Teachers' overwhelming workloads, due to large class sizes and numerous administrative responsibilities, often prevent them from giving individualized attention to each student

1.3 OBJECTIVES OF THE STUDY

General Objective

The general objective of this study is to develop a chatbot platform using Retrieval-Augmented Generation (RAG) to help students learn from academic lecture notes, while ensuring accuracy and credibility through content sourced directly from academic materials.

Specific Objectives

Develop a chatbot that supports student learning by integrating Retrieval-Augmented Generation (RAG) with a large language model (LLM) to provide accurate and contextually relevant answers sourced from trusted academic materials.

Develop an upload module that allows students to submit their lecture notes or study materials to the platform, enabling the chatbot to use the contents for accurate and personalized retrieval during interactions.

Enhance the Retrieval-Augmented Generation (RAG) pipeline to improve the accuracy and efficiency of capturing and retrieving relevant student notes.

1.4 SIGNIFICANCE OF THE STUDY

Theoretical Significance

The study contributes to the progression of AI integration in the education sector. By utilizing RAG with Large Language Models(LLMs) to provide reliable and contextually relevant answers, the study showcases how AI has the potential to revolutionize how learning resources are accessed and utilized. The findings from this research can help overcome the limitation of traditional Large Language Models(LLMs), particularly the issue of “hallucination,” and demonstrate a more robust framework for using AI to support learning. Lastly, this approach may pave the path for future advances in instructional technology and tailored learning.

Practical Significance

The study directly addresses a critical challenge in modern education: the gap between the information in lecture notes and a student's ability to effectively understand and retain it. By developing a Retrieval-Augmented Generation (RAG) chatbot, this research provides a practical and innovative solution that benefits key stakeholders and advances the field of educational technology.

The study will provide students with a tool for independent learning. By offering immediate, accurate, and context-specific answers drawn directly from their own lecture notes, the RAG chatbot will help students overcome common learning hurdles, such as fragmented information and difficulty in understanding complex concepts, learning to improve information retention and academic performance.

For instructors and academic institutions, this study provides a solution that improves efficiency and educational quality. The chatbot eases the strain on teachers by automating responses to common student questions, freeing up crucial time. This enables educators to focus on more important practical duties, such as creating compelling classes and providing in-depth support for challenging topics. The effective application of this technology can serve as an example for schools seeking to integrate AI-powered tools into a more efficient and modern learning environment.

1.5 SCOPE AND LIMITATION

Scope

The purpose of this study is to create a Retrieval-Augmented Generation (RAG) chatbot that will help students in asking enquires using academic lecture notes. The system's primary function is to allow students to upload digital lecture notes like PDF which are eventually processed into a knowledge base for retrieval and response generation. The chatbot uses a large language model (LLM) and a retrieval pipeline to generate responses that are grounded in the uploaded documents.

It covers the design, development, and testing of the chatbot as a prototype learning tool. It highlights basics such as document intake, knowledge retrieval, and natural language answer development, making it helpful in both the classroom and self-study settings. The system's target audience are students at a certain school who regularly use lecture notes as their primary learning resource.

The system developed in this study can handle file types such as pdf, docx, and text files. The system support multiple uploads simotaneously and

Functional testing and user-based assessment are the only methods used to evaluate the system. These methods focus on the chatbot's accuracy, efficiency in collecting information from lecture notes, and efficiency in assisting independent study. This limitation offers a proof-of-concept that illustrates the possibilities of RAG-powered chatbots in education, but it excludes extensive implementation and integration with learning management systems (LMS). Furthermore, even though the study's primary focus is on text-based lecture notes, other file formats—like pictures, handwritten notes, and multimedia—are not included and are instead being explored as potential areas for improvement.

Limitations

This study adheres to current privacy and ethical regulations, ensuring that all practices align with industry standards. A key limitation in this research is the use of academic notes and proprietary materials, which may raise concerns regarding intellectual property and privacy. Given the sensitivity of using such materials, the study deliberately avoids relying on academic notes or restrictions. Instead, it uses open-access lectures and publicly available educational content for demonstration purposes.

The Chabot developed in the study can only support PDF's, word, and text files. It cannot process files such as HTML, JSON, or domain specific file types. This limitation is due to the differences in the preprocessing logic of every file type. Additionally, the system currently lacks the capability to capture images if there exist in the documents, due to the system lacks of integration of multimodal tools such as Optical Image Recognition and Multimodal Large Language Models because of resource and budget constraints, though the logic in implementation of the system in this study can be easily adapted to support image recognition, which will be explained further in recommendations for future work.

This study was conducted within the constraints of available resources, including time, budget, and computational capacity. As a result, the implementation focuses on a feasible and manageable approach suitable for a prototype or proof-of-concept. While this allows for the demonstration of the core concepts, it may limit scalability, performance, or the inclusion of more advanced features. These aspects are acknowledged as potential directions for future work.

Another limitation of the platform developed in this study is that while it allows users to upload dynamic data, it is not fully equipped to control or monitor the types of content that users may upload. Although the platform is designed primarily as a chatbot for accessing lecture notes and educational resources, there is a risk that users might upload data or generate prompts that fall outside the intended scope of the system. The platform's open-ended nature means that users might misuse the system for generating unintended or inappropriate prompts, which could lead to responses that are not aligned with the educational objectives. For example, some users may upload irrelevant or non-educational materials or even attempt to prompt the system to produce content unrelated to the learning environment.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 RELATED LITERATURES

Challenges in Education

It was tackled complete literature review discussed on the difficulties in the implementation of the Special Education (SPED) curriculum in the Philippines. Even in the face of current legislation and policies to facilitate access to learning, schools continue to face obstacles in such areas as inadequate resources, insufficient teacher preparation, and policy practice discrepancy. Racism towards disabled students and family financial barriers are the main obstacles to constraints are barriers to effective special education implementation.

These are challenges that result in gaps in the provision of quality education, particularly in rural areas. According to the report, developing SPED services requires an increased government budget, effective teacher training, policy consistency, public awareness, and family support[13].

Cognitive load from extensive learning notes

The study about The Effect of Cognitive Load on Learning Memory in the accounting student in the Philippines emphasizes that intrinsic and extrinsic cognitive load must be reduced to enhance students' learning efficiency. This aligns with the idea that extended lecture materials, when not carefully designed, may overwhelm students' working memory. Thus, cognitive difficulties arise when learners struggle to process and store excessive or poorly structured lecture content.

The findings highlight that improving LMS quality and reducing extraneous load can directly support students' comprehension. This connects with challenges in understanding extended lecture notes, where clarity, pacing, and organization are essential for preventing overload. When lecture materials are fragmented or overly complex, students' working memory becomes strained, reducing comprehension.

The results also showed that satisfaction and improved e-learning quality enhance long-term memory retention. This indicates that properly managed instructional design can mitigate comprehension problems in lecture materials. In contrast, extended lecture notes without supportive structure or digital scaffolds may weaken both memory storage and motivation, further illustrating the difficulty students face in understanding them. (Sunga et al 2024) (local)

<https://www.mdpi.com/2071-1050/16/4/1686>

Learners interacting with digital lecture materials often encounter increased cognitive demands. Extended digital content requires sustained attention and the simultaneous processing of multiple elements, which can overwhelm working memory and reduce

comprehension efficiency. The cognitive load imposed by digital formats highlights the importance of designing learning materials that account for students' limited mental resources.

Digital environments can slow learners' processing and response times, particularly for complex or detailed information. When cognitive resources are stretched, students may rely on surface-level processing strategies, such as memorization, rather than engaging in deeper cognitive strategies like organizing, elaborating, or monitoring information. This can hinder the ability to fully understand and retain extended lecture materials delivered digitally.

These findings imply that digital lecture content must be carefully structured to minimize extraneous cognitive load and support meaningful learning. Incorporating strategies that promote attention management, reduce perceptual and structural disruptions, and encourage active engagement can help learners process and retain digital materials more effectively. Without such support, extended digital content may lead to persistent difficulties in comprehension and memory retention.

<https://link.springer.com/article/10.1007/s10639-025-13668-5?utm> (international)

The Role of AI and Machine Learning in Education

Artificial Intelligence (AI) is positively transforming higher education, reshaping how students learn and how teachers teach academics (Zouhaier, 2023). Instead of traditional methods, educational institutions are adopting these technologies to keep up with the innovation of our rapidly changing world (Tsou, 2023; Arthur D. Little, 2024). Important AI subfields like machine learning (ML) and natural language processing, which enable AI systems to evaluate student data and modify lessons to suit different learning styles and speeds, are at the center of this integration.

Better learning outcomes are achieved through individualized instruction, specialized tutoring programs, and quicker evaluation of student work thanks to this personalized approach. AI helps teachers at the same time by automating tedious chores and guaranteeing that students receive insightful and consistent feedback (Harry, 2023).

It discovers highlights on how artificial intelligence (AI) is improving education through providing personalized learning, developing grading systems, and helping teachers in providing more effective training. It emphasizes the benefits of AI, such as adaptive learning tools, automation grading, and insights based on data that improve teaching and learning. At the same time, it recognizes challenges such as ethical factors, data protection, usability problems and the need for educators to have enough training. Overall, the study focuses on AI's potential to improve education while emphasizing the significance of accessibility, transparency, and ethical responsibility[12].

Student learning through AI

In the Philippine educational context, the integration of AI offers a multitude of opportunities for enhancing the overall learning experience. AI-driven tools are increasingly seen as a way to augment educators, helping them better understand students' learning progress and tailor teaching strategies accordingly (Estrellado & Miranda, 2023). While the prospects of AI integration in the Philippine industry are promising, the progress is still in its early stages. Important issues such as privacy concerns, data security, and the potential for widening the digital divide are critical challenges that must be explicitly addressed (Holmes et al., 2021, as cited in Estrellado & Miranda, 2023).

While AI's impact on education is broad, its most significant contribution lies in directly supporting student learning by making it more personalized, engaging, and efficient. AI technologies like machine learning and natural language processing analyze student data to tailor the educational experience to each individual's pace and learning style (Harry, 2023). This leads to improved outcomes through personalized learning systems, intelligent tutoring, and automated grading, which save teachers time and provide consistent feedback.

Chatbots powered by Large Language Models (LLMs) in Student Learning

Chatbots that are powered by Large Language Models (LLMs) has changed the way students interact with learning materials by providing personalized on-demand support. These advanced AI technologies, such as OpenAI's GPT models, can improve student learning by responding instantly to questions, providing thorough explanations, and adjusting to different learning styles. This individualized learning experience is especially effective in increasing engagement since students can get quick help with homework, clarify concepts, or explore topics further without having to wait for a teacher's availability (Ayala & Béchar, 2024). The ability to receive immediate feedback and support enhances learning and keeps students engaged, especially in self-paced situations where they can study at their own pace.

Some of the main features of LLM-powered chatbots is their capacity to give specific learning experiences. These chatbots assess students' questions and adjust responses to their level of knowledge, ensuring that explanations are clear and relevant. It proves particularly helpful for several classrooms with students of different degree of competency in different topics (Gupta et al., 2024). Chatbots can assist students narrow learning gaps and provide additional support by providing guidance that is customized to their requirements. Furthermore, the ability of these chatbots to handle repeated questions or commonly requested topics frees up significant time for instructors to focus on advanced parts of education, therefore enhancing the entire learning experience.

However, with their potential, LLM-powered chatbots are limited. They can also generate wrong or incomplete responses, a condition known as hallucination, in which the chatbot offers answers that seem logical but are inaccurate in fact (Lewis et al., 2020). This can be particularly challenging in educational environments when accuracy is critical to student learning. Furthermore, while chatbots thrive at presenting factual information, they may struggle with higher-order cognitive tasks like encouraging critical

analysis or facilitating deeper conversations. As a result, while they can be an effective additional tool, they should be used along with traditional teaching approaches to ensure that students receive full and reliable information.

Despite these challenges, chatbots in education provide substantial benefits, particularly in terms of active learning. Students can interact with chatbots by asking follow-up questions and receiving personalized explanations, which improves retention and understanding. In addition, the usage of chatbots promotes self-directed learning by allowing students to revisit topics, test their knowledge, and receive quick feedback, thus supporting continual learning and development. Chatbots act as both tutors and learning companions, assisting students in achieving their academic goals more successfully.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has emerged as a solution to mitigate some of the challenges faced by LLMs, particularly hallucination and limited knowledge generalization (Lewis et al, 2020; Ayala & Béchar, 2024; Kirchenbauer & Barns, 2024). RAG is an architecture designed to enhance LLMs by first searching for useful information before responding, rather than answering solely based on the LLM's learned knowledge from training. Large Language Models (LLMs) are trained on vast volumes of data and use billions of parameters to generate original output for tasks like answering questions, translating languages, and completing sentences. RAG can extend the already powerful capabilities of LLMs by grounding its responses in specific domains or an organization's internal knowledge base without retraining the model (Gupta et al., 2024). RAG models can ground outputs in reliable documents, ensuring responses are both relevant and verifiable, which could be beneficial for dealing with educational materials.

Fig. 1 shows the basic process of how LLMs work. When a user submits a query, the model processes the input and generates a response. The input can be any text, like a question or statement. The LLM (such as GPT or LLaMA) starts by breaking the input into smaller parts called tokens, which represent words or subwords. Next, the model uses a mechanism called self-attention to understand how these tokens relate to each other. This helps the model understand the meaning of the query. After processing the input, the model predicts the next word or phrase based on what it has learned. It generates the response step-by-step, using the previous words to predict what comes

next. Finally, the model displays the generated text as its response, ensuring it is relevant and coherent based on the input.

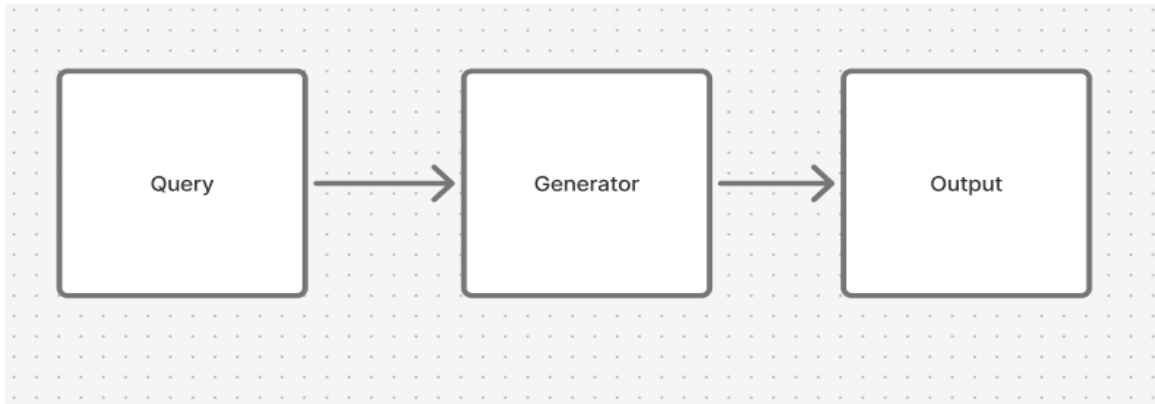


Figure 2 illustrates how Retrieval-Augmented Generation (RAG) enhances the functionality of LLMs. The approach can be broken down into three main stages:

Indexing: In this stage documents into chunks and uses an ending model to turn those chunk into a vector along with its text representation and metadata. Each of these are then stored into a vector database to where the retriever searcher for relevant information.

Retrieval: Before generating a response, RAG searches an external knowledge base (e.g., PDFs, DOCX files, or web content). The input query and relevant documents are vectorized, meaning they are converted into numerical representations that capture the semantic meaning of the content. Typically, vectorization techniques like BERT embeddings or Sentence-BERT are used for creating these vectors. The vectorized documents are then stored in a vector database for efficient similarity-based retrieval. The most relevant documents are retrieved

Generation: The most relevant documents retrieved during the retrieval phase are used to provide context for the LLM. These documents, combined with the model's learned knowledge, inform the generation of a response. This ensures that the model's output is not only based on the data it was trained on but also grounded in specific, relevant

information retrieved from external sources, thereby enhancing the accuracy and relevance of the response. However, aspects like tone, fluency, and coherence are still primarily handled by the LLM's pre-existing knowledge, as the model generates the response based on both the retrieved context and its trained language capabilities.

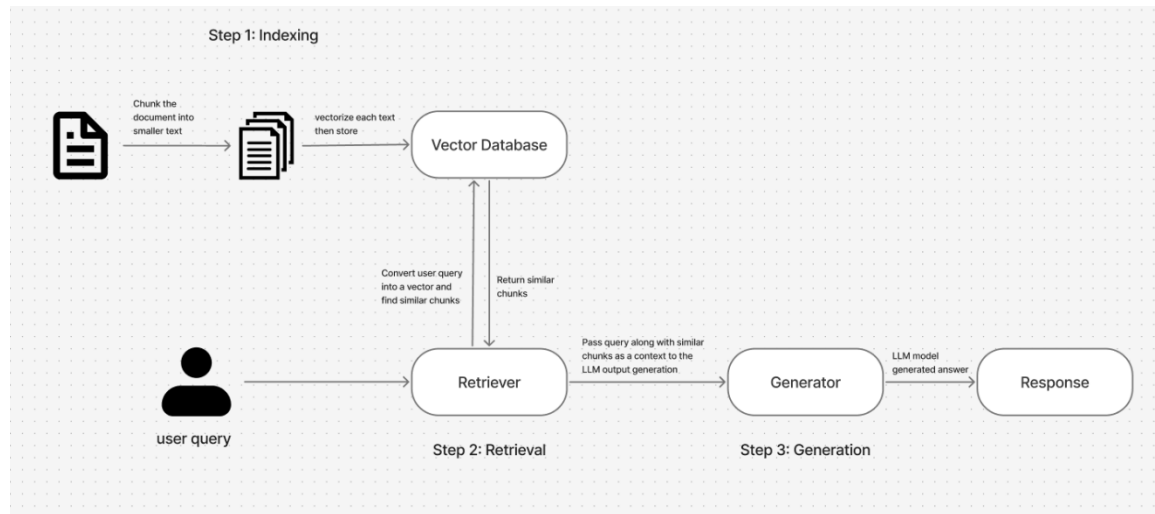


FIG 2. Basic RAG Process

React and Fast API as an app development stack

React is a declarative, component-based library that allows developers to build reusable UI components. Its modular architecture enables the creation of highly interactive and responsive user interfaces that update dynamically as data changes, while preserving a clear separation of concerns. These characteristics make React particularly well-suited for applications where an engaging user experience is a priority (React, n.d.).

On the server side, FastAPI is a modern, high-performance Python web framework built on Starlette and Pydantic. It is designed to maximize both speed and developer productivity by providing features such as automatic data validation, type annotations, and interactive API documentation through OpenAPI and Swagger. Its asynchronous architecture enables the handling of many concurrent requests, making it particularly suitable for real-time applications and microservices. Moreover, using FastAPI as a backend solution facilitates seamless integration with the extensive ecosystem of Python machine learning libraries, allowing developers to incorporate advanced data processing and AI capabilities directly into their applications (FastAPI, n.d.).

When combined, React and FastAPI provide a robust stack for modern app development: React manages the frontend presentation and interactivity, while FastAPI powers the backend logic, APIs, and data processing. This separation of concerns promotes scalability, maintainability, and rapid development, making the stack particularly effective for building data-driven applications such as AI-powered chatbot applications.

2.2 RELATED STUDIES

Enhancing student learning through Generative AI

Recent research in Metro Manila shows how students are beginning to engage with generative AI technologies in their academic work. Arguson, Paculan, and Mabborang (2024) found that while students are generally open to using AI for homework, their acceptance largely depends on their attitudes toward the tools, their confidence in using them, and the accessibility of these systems. Among AI platforms, ChatGPT was the most familiar and widely used compared to Copilot and Gemini. These findings suggest that the Philippine education system is not only facing legal and ethical challenges but is also experiencing a cultural shift as students increasingly view AI as a learning partner (Arguson, Paculan, & Mabborang, 2024).

A separate study looked at how instructors in community colleges in Albay, Philippines, are incorporating AI into student performance assessments. The research highlights several benefits, including opportunities for personalized learning and improved evaluation, but it also points out challenges such as limited facilities, lack of training, and ethical concerns about privacy and fairness. Results indicate that while both students and educators recognize AI's potential, there are still significant gaps in confidence, equal access, and ethical awareness—especially among part-time instructors. To address these issues, the study proposes an AI framework focused on monitoring, operational security, and instructional support to ensure ethical and effective AI integration in education (Tabuena, 2025).

Similarly, the article *Generative AI and Education: Dynamic Personalization of Pupils' School Learning Material with ChatGPT* explores how ChatGPT-3.5 and 4 can be used in primary education to personalize lessons. In a study of 110 Uruguayan pupils aged 8–14,

generative AI was used to adapt history lessons—texts, illustrations, and exercises—based on students’ knowledge levels. The results showed that AI-driven personalization improved motivation, engagement, and performance by supporting intrinsic motivation and creating a sense of “flow” in learning. Still, the study cautions against overreliance on AI and raises concerns about data privacy and accessibility. Overall, the findings suggest that, when used thoughtfully, generative AI can significantly enhance pupils’ learning experiences (Jauhiainen & Garagorry Guerra, 2024).

Meanwhile, the article *Enhancing Student Engagement Through Artificial Intelligence (AI): Understanding the Basics, Opportunities, and Challenges* introduces the core concepts of AI and examines its role in reshaping higher education. It explains how tools such as generative AI and chatbots can personalize instruction, provide adaptive feedback, and encourage interactive learning across different contexts. The authors also emphasize critical concerns, including ethical issues, data privacy, algorithmic bias, and risks to academic integrity. The paper concludes by calling for more research on adaptive systems, ethical AI practices, and collaborative learning models to maximize the benefits of AI while ensuring fairness and student well-being (Nguyen, Kremantzis, Essien, Petrounias, & Hosseini, 2024).

Hallucination in LLMs

Recent Large Language Models (LLMs) have exponentially improved in fluency due to the development of new solutions such as transformer-based architectures. However, attention towards their limitations and potential risks has also increased (Holtzman et al, 2020). LLM models are particularly notorious in the generation of outputs that are coherent and plausible but factually incorrect or misleading. This phenomenon, known as "hallucination," has been a growing focus of research (Huang et al, 2024). These inaccuracies pose significant issues in educational settings, as students might inadvertently accept misleading information, resulting in incorrect conceptual understanding and lasting misconceptions. This is particularly relevant for students with limited subject matter expertise. They are especially vulnerable to overreliance on AI-generated responses. Without the skills to independently verify information, they may

accept erroneous outputs at face value, compounding the problem of misinformation in learning environments (Zhai et al, 2024)

The issues with LLMs like ChatGPT often produce responses interspersed with inaccuracies or fabrications (Khalil & Er, 2023). A study by Gao et al. (2023) proved this issue by utilizing 50 abstracts sourced from five scientific journals and assigning ChatGPT to create abstracts based on their titles. Subsequently, both sets of abstracts underwent scrutiny by AI plagiarism detection software and impartial human assessors. The results revealed that out of the abstracts generated by ChatGPT, 68% were correctly identified as such (true positives), while 14% of authentic abstracts were mistakenly categorized as chatbot-generated (false positives). These findings underscore the model's limitations in accurately generating information, which can hinder its ability to provide reliable sources for users. Another study is by Watts et al. (2023) wherein the authors evaluated the mechanistic reasoning of three dialogue systems (ChatGPT-3.5, ChatGPT-4, and Bard). The findings revealed that chatbot responses either underperform or reason on par with students and occasionally provide inaccurate answers to content questions. The results suggest that dialogue systems exhibit limited mechanistic reasoning capabilities compared to students. This study suggests that students, when equipped with knowledge, are capable of questioning the reasoning of LLM models, as these models sometimes provide inaccurate answers.

Hallucinations occur because of how the models are trained. Language models first learn through pretraining, a process of predicting the next word in huge amounts of text. Unlike traditional machine learning problems, there are no “true/false” labels attached to each statement. The model sees only positive examples of fluent language and must approximate the overall distribution. This may generate plausible but incorrect information that fits learned patterns. Second, Models are trained on large static data, which may contain inaccurate data or be outdated in time. Third, there is an unequal distribution of what can be used to train the model. Not all knowledge fields can be equally represented, such as topics that are complex, such as flight dynamics or fluid mechanics, where domain-specific data may be scarce or poorly represented in public datasets. And lastly, there is an upper bound knowledge capacity of models. Models use computational power to accommodate room for more learning. However, there is no model yet powerful enough

to fully encode all human knowledge, especially for rare or domain-specific facts. All these can be attributed to why hallucinations persist, and future research may target these factors to minimize hallucinations from LLMs (Ji et al., 2024; Kalai et al., 2025).

Rag in mitigation of Hallucinations

The utilization of RAG for reducing hallucinations has seen success, researchers conducted studies on rag utilization for mitigating hallucinations, by simply integrating an external knowledge base for retrieval and observing outputs, they have recorded up to 60% reduction in hallucinated responses, demonstrating improved factual accuracy in dialogue systems and better handling of knowledge-grounded conversations, the chatbot generates more reliable and contextually relevant answers, mitigating the risks of generating plausible but incorrect information. Through these finding it demonstrated how RAG can effectively combat hallucinations and improve the reliability of AI-generated responses which is crucial for building applications focusing on precise information (Bécharde et al., 2024 : Shuster et al., 2021) .

However RAG models can also suffer from hallucinations. Fig 3 identifies some of the common Causes and possible solutions provided by researchers , In this research of () they stated that Rag Hallucinations causes can be separated into two primary stages: retrieval failure and generation deficiency.:

Retrieval Failure

- **Query Problem:** The initial query is not effective at retrieving the right information.
 - **Query Refinement:** Improving the original query with techniques like rephrasing or adding context.

- **Query Decomposition:** Breaking a complex query into smaller, more manageable sub-queries.
- **Query Decontextualization:** Isolating the core of the query by removing irrelevant conversational context.
- **Retrieval Strategy Problem:** The method used to retrieve information is flawed.
 - **Adaptive Retrieval:** The model dynamically adjusts its retrieval strategy based on the query's needs.
 - **Iterative Retrieval:** The model retrieves information in multiple steps, refining its search with each iteration.
 - **Recursive Retrieval:** The model uses the results of one retrieval to inform and generate subsequent queries.
- **Retriever Problem:** The retriever itself, the component responsible for finding information, is not performing well.
 - **Search Granularity:** The size of the retrieved text chunks is either too large or too small, leading to poor results.
 - **Embedding:** The vector representations of the queries and documents are not accurate enough to match them effectively.

Generation Deficiency

- **Context Noise:** The retrieved context contains irrelevant or distracting information.
 - **Re-Ranking:** Sorting the retrieved documents to prioritize the most relevant ones.
 - **Context Compression:** Using techniques to remove extraneous or redundant information from the retrieved text.

- **Irrelevant Context Training:** Training the model to ignore or filter out irrelevant data during the generation phase.
- **Middle Curse:** The LLM prioritizes information at the beginning and end of the retrieved context, ignoring crucial details in the middle.
 - **Middle Priority:** use techniques that explicitly increase the model's attention to the middle part of the context or just move the relevant context in the middle.
- **Alignment Problem:** The generated response doesn't align with the retrieved context or the original query. This is a core issue where the model's internal knowledge conflicts with the provided information.
 - **Context Faithfulness:** Ensuring the output generated is directly supported by the retrieved context.
 - **Query Faithfulness:** Ensuring the response directly and accurately answers the original user query.

Hallucinations in Retrieval-Augmented Generation (RAG) models stem from multiple causes, and various techniques have been proposed to address them. Mitigating these issues is critical, as the very purpose of RAG is to ground generative outputs in an external knowledge base to improve factual accuracy. If RAG itself produces hallucinations, however, its fundamental advantage is undermined, calling into question its effectiveness as a reliable framework for reducing errors in large language models.

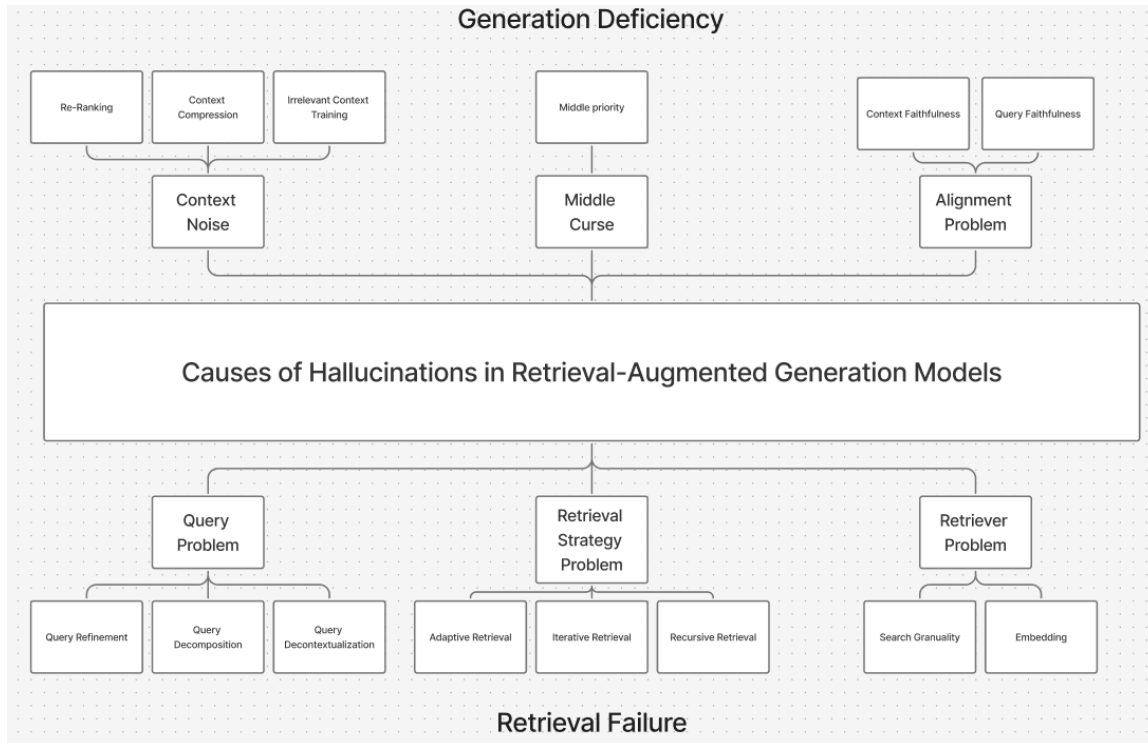


Fig 3 RAG Identified Hallucination Causes and possible Solutions

Retrieval-Augmented Generation (RAG) models can be implemented in a variety of ways, with their pipelines typically optimized according to the type of data being processed and the specific requirements of the task. To address the persistent issue of hallucinations, several researchers have proposed novel methods aimed at improving the reliability of RAG outputs.

One notable approach is RAG-HAT, introduced by [Author(s), Year], which employs a fine-tuning pipeline centered on hallucination detection and mitigation. This framework integrates hallucination detection mechanisms with corrective strategies, allowing the model to adaptively identify and suppress hallucinated content during the generation process. Empirical results demonstrated that RAG-HAT significantly reduced hallucinations and enhanced the performance of large language models (LLMs) in real-world tasks. In contrast to fine-tuning methods, [Author(s), Year] proposed ReDeEP, a detection-focused framework that leverages mechanistic interpretability to understand how LLMs utilize both external

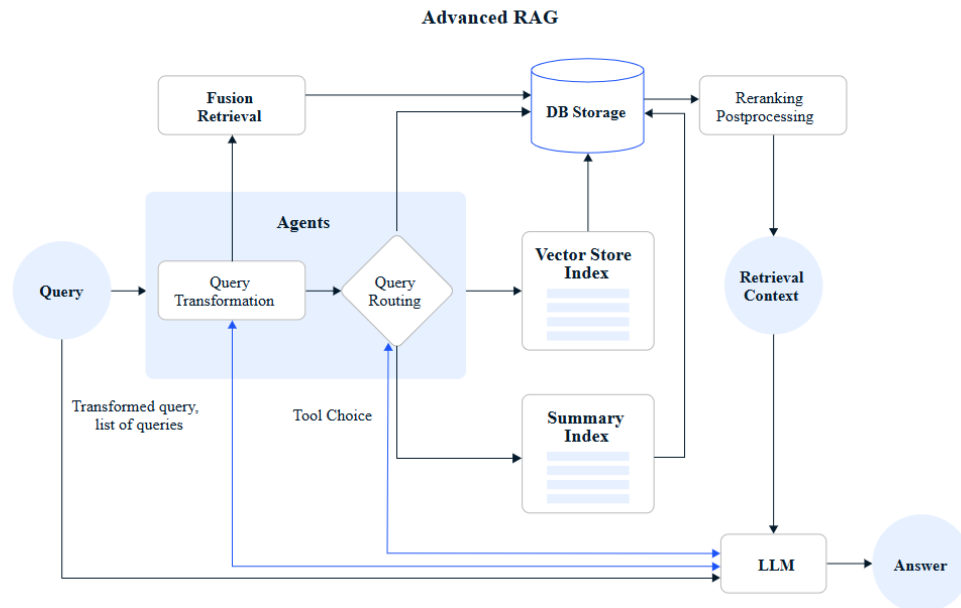
and parametric knowledge. By providing insights into when and why hallucinations occur, ReDeEP advances detection accuracy and offers a foundation for more targeted error identification and mitigation strategies. A further advancement is presented by [Author(s), Year] in the form of LRP4RAG, which applies Layer-wise Relevance Propagation (LRP) to track the relevance of input components during the generation process. This approach enables more precise detection of hallucinations by analyzing the contribution of each input element. Experimental findings confirmed that LRP4RAG outperforms existing methods, offering improved reliability and trustworthiness of RAG-generated responses across diverse contexts.

Collectively, these studies underscore the evolving landscape of hallucination mitigation strategies in RAG pipelines, ranging from fine-tuning approaches to interpretability-driven detection methods. Such advancements contribute to enhancing the factual accuracy and dependability of LLMs, marking significant progress toward addressing the issue of hallucinations..

Advances in RAG

Multiple researchers have discussed different ways to improve RAG in both the retrieval and generation phases, due to the vast efforts in implementing rag we have structured it, into different phases

The shift towards advanced RAG



Advanced RAG: Architecture, techniques, applications and use cases and development (LeewayHertz, 2025) The shift towards advanced RAG, for this the Advanced RAG systems have emerged to overcome the limitations of basic models by integrating more sophisticated techniques. They use semantic search and contextual understanding to improve retrieval, while also incorporating feedback loops and dynamic updates to keep knowledge accurate and current. These systems enhance the four main components—storing, retrieving, augmenting, and generating—through semantic indexing, contextual retrieval, adaptive learning, and more coherent response generation. Overall, the shift to advanced RAG marks a major step forward, enabling more accurate, personalized, and contextually aware AI-driven interaction.

Advanced RAG systems in enterprises rely on a well-designed architecture composed of several essential parts: data preparation/management, user input processing, retrieval, information generation, feedback loops, integrations, and security. Data is cleaned, chunked, vectorized, and enriched with metadata; user queries are rewritten, guarded, and contextualized; retrieval involves indexing, re-ranking, hyperparameter tuning, and handling complex formats. The generation module not only composes responses using powerful LLMs but also applies moderation, caching, personalization, and monitoring to ensure responses are safe, efficient, and tailored. Finally, feedback and continuous

improvement cycles—along with enterprise integrations (APIs, CRM/ERP systems), scalability, and strong security/compliance measures—ensure the RAG system stays effective, relevant, and trustworthy over time.

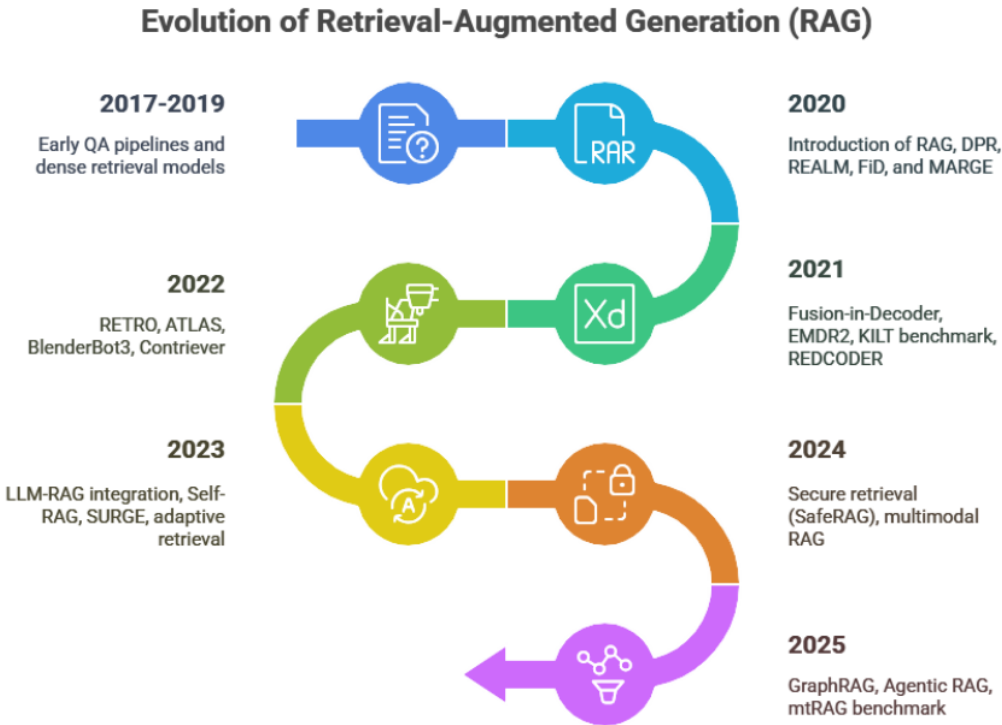


Figure 2 Oche, Folashade, Ghosal, & Biswas, 2025), Retrieval-Augmented Generation (RAG) started with early QA pipelines and dense retrieval models (2017–2019), then officially launched in 2020 with RAG, DPR, REALM, and FiD. In 2021, better fusion methods like Fusion-in-Decoder and benchmarks like KILT improved evaluation. By 2022 and 2023, models like RETRO, ATLAS, and Self-RAG showed that retrieval can replace huge model sizes and make chatbots more accurate. Recent work (2024–2025) focuses on secure, multimodal, and agent-based RAG, aiming for smarter, safer, and more reliable AI systems.

Challenges and Limitations of RAG

Despite the numerous virtues of incorporating RAG, several challenges need attention. The retrieval mechanism, which is helpful for grounding answers in relevant information, can still struggle, particularly when dealing with ambiguous queries or niche knowledge domains. The Dense retrieval reliance on dense vector representation can sometimes lead to irrelevant or off-topic documents being retrieved. Efforts to refine retrieval techniques, including the incorporation of more sophisticated query expansion and contextual disambiguation, are needed to improve performance in these areas. While the integration of both retrieval and generation is seamless in theory, in practice, it may break. The generative module may not effectively incorporate the relevant information in its response. Research in enhancing the generation to have a better alignment mechanism, such as improved attention models or hierarchical fusion techniques, may help alleviate these issues.

In the generation mechanism, it may face issues in flawed and inadequate data, which can confuse the model into producing responses that are misleading, incomplete, or contextually off-target. This limitation highlights the importance of ensuring the quality and relevance of the retrieved information and the effectiveness of the augmentation process by cleaning, filtering, and verifying to ensure the integrity and reliability of the retrieved information. LLM models also have a token allowance. Meaning there is a limit on the number of tokens it can process, tokens may vary in the input and output of the model, and a proper RAG should respect this boundary to ensure the generation is complete. There is also a challenge in structuring the retrieved information that is to be presented to LLM. LLM may miss out on extracting the most relevant information from long structures of information. To address this issue, advanced RAG systems can employ techniques such as `_diversity-aware` ranking or information salience detection. By prioritizing and ordering the retrieved examples based on their relevance, diversity, and informative value, RAG systems can ensure a more balanced and effective presentation of the information to the LLM.

RAG systems can introduce additional latency as rag involves several steps. This limitation can be particularly challenging in real-time or interactive scenarios where quick response times are critical. To mitigate this issue, RAG developers can look into caching,

pre-computing, and parallel-processing strategies to reduce the frequency of processing similar information and to and to leverage the power of parallelism in today's computers.

Rag applications in education:

Studies show that RAG-based systems give more accurate and specific answers in many tasks. They also keep knowledge updated by pulling information from the latest sources during inference, which helps LLMs stay current without expensive retraining (Gao et al., 2023; Shuster et al., 2021). In education, researchers found that RAG chatbots reduce hallucinations, support student learning, and improve access to school information, making them useful for academic support (Swacha & Gracel, 2025).

Husain, Wibisono, and Anisyah (2025) created a RAG-based academic chatbot to improve information services in higher education. Unlike regular chatbots and LLMs that often make mistakes or “hallucinate,” this chatbot uses data from official knowledge bases before answering (Ji et al., 2023; Lewis et al., 2020). Tests with 100 questions showed strong accuracy for factual answers, with a faithfulness score of 0.91, though it was weaker in reasoning tasks. It also rejected irrelevant questions 81.25% of the time (Gao et al., 2023). These results suggest RAG chatbots can be trusted in universities, but they still need improvements in reasoning (Wu et al., 2024).

Another study introduced the BiWi AI Tutor, a RAG-based chatbot made to give scalable, personalized, and context-aware mentoring in higher education. It combines an LLM with indexed course materials, allowing it to give accurate answers to both academic and school-related questions. Results showed up to 87% accuracy, with GPT-4 evaluations matching well with human experts. Accuracy improved further with reranking, especially for school-related questions. The study shows that RAG chatbots can improve digital learning, though more work is needed in personalization, ethics, and mentoring-style features.

MARK system (Machine Assistant with Reliable Knowledge) was also developed as a RAG-based assistant to give students accurate, context-based answers while reducing hallucinations. It uses a mix of keyword, sparse, and dense retrieval with reranking to make sure answers are reliable and tied to course content. Tested as a virtual

tutor in engineering courses and as a tech support tool, MARK handled both academic and service-related questions well and improved through feedback. Still, it depends on the quality of its knowledge base and currently only supports text, showing the need for future upgrades like multimodal features and deeper personalization.

2.3 SYNTHESIS

AI has been continually being adopted as tools for helping student learning , by leveraging AI innovative tools has shown promise aiding the

CHAPTER 3

RESEARCH DESIGN AND METHODOLOGY

Introduction

Theoretical Framework

This study is anchored on theories and models in the fields of Information Retrieval and Representation Learning, which together provide the foundation for the Retrieval-Augmented Generation (RAG) paradigm.

First, Information Retrieval (IR) Theory explains the process of locating relevant information from large document collections. Classical IR models, such as the Vector Space Model (Salton, 1975) and the Probabilistic Model (Robertson & Sparck Jones, 1976), established how documents and queries can be represented mathematically, and how similarity measures (e.g., cosine similarity, BM25) can rank documents based on relevance. These theories support the role of the retriever component in RAG, where the system identifies the most relevant passages to a user query.

Second, the study draws from Representation Learning Theory in Natural Language Processing (NLP), which underpins modern dense retrieval methods. Word embeddings (Mikolov et al., 2013), contextual embeddings (Devlin et al., 2019 with BERT), and sentence transformers demonstrate how language can be mapped into high-dimensional vector spaces where semantic similarity can be captured beyond mere keyword matching. This theoretical foundation justifies the use of dense encoders in RAG systems for retrieving semantically relevant documents.

Finally, Generative Language Model Theory (Vaswani et al., 2017; Brown et al., 2020) explains how large-scale transformer models can generate coherent and contextually appropriate responses. When combined with retrieval mechanisms, these models overcome the limitations of closed-book generation by grounding responses in external knowledge. Thus, RAG embodies a theoretical synthesis of retrieval-based systems (which ensure factual grounding) and generation-based systems (which ensure fluency and flexibility).

Conceptual Framework

The Conceptual Framework of this study uses an Input–Process–Output (IPO) model to depict the logical flow of the research. The input stage includes the Query of the user which are questions about the lecture notes and then the Documents uploaded in the system. This data serves as the basis for analysis. The Process involves the indexing of the documents and query, retrieval using sparse, dense, and hybrid which will be reranked using a cross encoder. To be submitted to the Generation process which first assembles the prompt before using a Large Language Model to Generate the Final Output. And getting the retrieved documents data for reference to display on the system.



Research design

The study adopts a Mixed Methods Research that focuses on the development and implementation of a lecture note learning assistant based on Retrieval-Augmented Generation. This platform utilizes Information Retrieval along with Large Language Model(LLM). The focus is on generating answers based on a user's query that is grounded on the information stored on the system. The primary focus is to assess how Information Retrieval along with Generation using LLMs can improve the accuracy, reliability, and usefulness of responses to student inquiries. Data for evaluation will be gathered through testing the rag system on free lecture notes and the qualitative feedback of participants, The study focuses on evaluating both the system's ability to provide source-accurate information and students' perceptions of its effectiveness in supporting learning.

System Architecture

System Description

Frontend (User Interface) – The user interface is built using React, chosen for its component-based structure and fast development cycle. React allows us to design a responsive and dynamic interface where users can submit queries, view retrieved documents and interact with generated answers seamlessly. It also integrates well with backend APIs and supports state management libraries that improve user experience during asynchronous operations. Additionally, React's ecosystem provides flexibility for future enhancements, such as integrating dashboards, visual analytics, or real-time feedback collection.

Backend(Processing and Data Management)- *The backend of the system is developed using **FastAPI**, a modern Python web framework designed for building high-performance APIs. It serves as the central orchestration layer that connects the user interface, the retrieval-augmented generation (RAG) layer, and the knowledge stores. FastAPI was chosen because of its asynchronous request handling, automatic data validation, and compatibility with machine learning workflows, which make it efficient for handling multiple queries simultaneously. To run the FastAPI application, the system uses **Uvicorn**, a lightning-fast ASGI server built for asynchronous Python frameworks. Uvicorn handles incoming HTTP requests, manages connections, and ensures low-latency communication between the client and the backend. Its performance and seamless integration with FastAPI make it a production-ready choice for deploying the system.*

Data Stores- *The data stores are a compilation of databases that maintain structured data such as user profiles, authentication records, and system logs. For unstructured and semantic search, a **vector database** built with **FAISS** stores embeddings of documents, enabling fast and accurate similarity search.*

RAG Layer (AI Integration) – The RAG layer serves as the AI logic of the system and consists of three interconnected modules: indexing, retrieval, and generation. The **indexing module** converts documents into vector embeddings and stores them in FAISS for later access. The **retrieval module** fetches the most relevant passages from the knowledge store based on user queries. Finally, the **generation module**, powered by the large language model (LLM), synthesizes the final answer by combining the user's query with the retrieved context. This layered approach ensures both efficiency and factual grounding in responses..

Security – Security is enforced through JWT (JSON Web Token) Authentication, ensuring that all HTTP requests from the frontend are verified and authorized using unique tokens. This prevents unauthorized access to system resources and user data . information such as authentication tokens and user credentials are encrypted at rest, further safeguarding the system from data breaches.

System Tools Chosen

- **React.js**

React.js was selected for building the **frontend** of the system because it is a widely used JavaScript library that allows the creation of **interactive, component-based user interfaces**. Its virtual DOM and efficient rendering make it suitable for responsive applications where query input and real-time output streaming are important.

FastAPI

FastAPI was chosen for the **backend** implementation because it is a **high-performance Python web framework** designed for building APIs. It provides **asynchronous request handling, automatic validation, and easy integration with machine learning models**. This makes it particularly effective for orchestrating the retrieval-augmented generation (RAG) workflow, handling multiple user queries concurrently, and connecting with both the knowledge stores and AI components.

Uvicorn

To serve the FastAPI application, the system uses **Uvicorn**, a lightning-fast ASGI (Asynchronous Server Gateway Interface) server. Uvicorn is responsible for handling incoming HTTP requests, managing concurrent client connections, and ensuring low-latency communication. Its tight integration with FastAPI and high throughput make it a production-ready server for real-time AI applications.

- **LangChain**

LangChain was chosen as the LLM application development framework because

it provides a **rich API** that abstracts much of the complexity involved in building LLM-powered systems. It simplifies **prompt management, chaining of retrieval and generation steps, and integration with vector databases**, which makes it ideal for implementing a Retrieval-Augmented Generation (RAG) pipeline.

- **FAISS (Facebook AI Similarity Search)**

FAISS was selected as the **vector database** because it is optimized for **fast similarity search and clustering of high-dimensional embeddings**. It can efficiently handle millions of vectors, supporting both exact and approximate nearest neighbor searches. This makes retrieval of semantically relevant document chunks both **scalable and efficient**.

- **MiniLM v6**

MiniLM v6 was chosen as the **embedding model** due to its balance of **lightweight architecture and strong performance**. It generates high-quality embeddings that capture semantic meaning while being computationally efficient, making it practical for real-time retrieval tasks without requiring large GPU resources.

BAAI/bge-reranker-base

To enhance retrieval precision, the system incorporates the **BAAI/bge-reranker-base** model as a reranker. While FAISS retrieves a broad set of relevant candidate chunks, the reranker assigns finer-grained relevance scores and reorders results so that the most contextually appropriate passages are prioritized. This ensures higher-quality inputs to the generation step, improving overall system accuracy.

- **Mistral-Instruct**

Mistral-Instruct, a **fine-tuned large language model for instruction following**, was selected as the **generator** because it produces **coherent, instruction-aligned responses**. Its fine-tuning ensures it can follow user queries closely, making the system's answers more accurate, grounded, and useful.

Frontend Development

Authentication Modal

The students are required to log in or sign up to enable document uploads and chatbot use. The purpose of login is to allocate resources of the file server to authenticated students. When new students sign up, their login credentials are stored securely in the database using a reliable password hashing algorithm, and they are issued a web token to begin using the application securely. The web token, typically implemented as a JSON

Web Token (JWT), is attached to each subsequent request to verify the student's identity and maintain a secure session without repeatedly exposing credentials. Session management is handled by validating these tokens on the server side, ensuring that only authenticated users can access protected features such as document storage, personalized retrieval, and chatbot interactions. In addition, token expiration and refresh mechanisms are enforced to prevent misuse, while encrypted communication via HTTPS safeguards data transmitted between the client and server. Together, these measures ensure that authentication and authorization processes remain both efficient and secure for all users of the system.

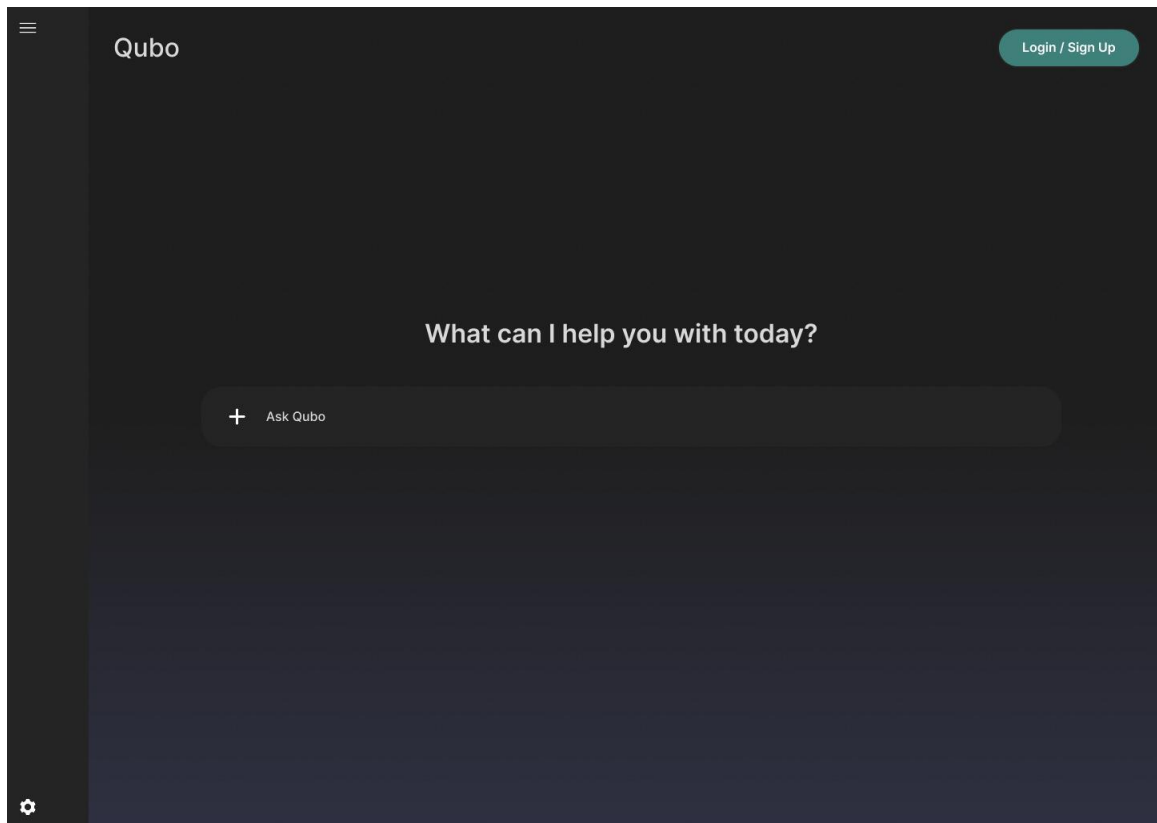
File Manager

The File Manager is a dedicated interface available only to authenticated students, providing secure space for managing their documents within the system. Through this component, students can view, upload, and organize files, ensuring that their personal or course-related materials are readily accessible. The File Manager supports common document formats such as PDF, DOCX, and plain text files.

The files uploaded to the File Manager will be the documents processed by the backend to store in the knowledge base and into the **RAG (Retrieval-Augmented Generation) layer**, where the retriever selects the most relevant passages in response to a student's query. The selected content is then passed to the **chatbot's generation module**, enabling it to produce accurate, context-aware answers that are grounded in the uploaded materials

Chatbot Interface

The student interface allows students or customers to submit natural language queries. As shown in Figure 1, Users enter their questions into a text input field and submit them using a designated button. Upon submission, the frontend issues a secure API request to the backend service. The backend processes the query, performs retrieval and generation operations, and returns a response via an encrypted connection. The frontend then renders the result within the interface, providing a seamless question-answering experience.



Backend Development

The backend of the system is implemented using Fast API, for its speed, scalability, and ease of integration with AI components. It serves as the bridge between the frontend and the RAG layer, exposing well-defined APIs that allow secure communication and efficient coordination across modules.

To ensure maintainability, the backend is structured around a separation of concerns, where different routes handle specific subtasks of the system.

- Authentication routes manage login, signup, token issuance, and session validation.
- File management routes handle document upload and storage operations.

- Retrieval and generation routes connect with the RAG layer, enabling the chatbot to process queries, retrieve relevant documents, and return generated responses.

This modular design allows the backend to act as a central controller: orchestrating document ingestion into the knowledge base, coordinating retrieval and reranking, and delivering results back to the frontend UI components (chatbot, file manager, and authentication modal). The use of Fast API also ensures that the system can support asynchronous requests, making it highly responsive even under concurrent student interactions.

RAG Layer Development

The RAG Layer is divided into three crucial components, indexing, retrieval, and generation Modules.

Indexing Module

The Indexing Module handles the file preprocessing needed for Retrieval. Later on, it includes the processes of extraction of the files content, Chunking, Embedding, then the index store

File Extraction

The contents of the documents are extracted, turning them from pdf's to a text file, docx files are first converted into pdf's . The extraction is performed using a tool called pdfplumber , which is preferred because it provides more accurate parsing of text compared to many alternatives. Unlike simple PDF-to-text converters, pdfplumber can:

- Preserve the logical flow of the document, including columns and spacing.
- Extract structured elements such as tables, headers, and footers with higher accuracy.

- Avoid unnecessary OCR when text is already embedded, ensuring faster and cleaner results.
- Handle complex layouts better, reducing data loss or misalignment.

This makes it particularly reliable when working with PDFs that contain both plain text and structured content

Semantic Chunking

Extracted documents usually is too long, which might cause the retriever to fetch whole corpora which is unnecessary and intensive, to handle this, a chunking strategy is applied. Chunking is the process of dividing long documents into smaller segments, commonly referred to as *chunks*. Each chunk typically contains only a few hundred words, allowing the retrieval system to search more precisely and return only the most relevant segments in response to a query. A chunking strategy is crucial to keep relevant sections together and ensure it doesn't cut mid sentences. Traditional chunking approaches use fixed counts of either word, sentence, or paragraph to determine the length of chunks. While this ensures that the text is manageable for retrieval and processing, it often produces arbitrary cuts that fail to align with the semantic flow of the document. For example, a definition may be split across two chunks, or an example may be separated from the concept it explains. Such breaks reduce retrieval accuracy and increase the likelihood of fragmented or incomplete responses.

A strategy that is gaining popularity is semantic chunking wherein instead of cutting in fixed lengths divides text into coherent units of meaning, ensuring that each chunk contains a logically consistent idea. Instead of relying on static boundaries, semantic chunking leverages embeddings which are mathematical representations of sentences or paragraphs to measure semantic similarity. The process begins by splitting a document into sentences. Each sentence is then transformed into an embedding vector using a pre-trained sentence transformer MiniLM v6 a lightweight yet powerful transformer-based model that achieves performance comparable to larger models (such as BERT or RoBERTa) while requiring significantly fewer computational resources. The

v6 variant is a fine-tuned version of MiniLM, optimized specifically for semantic textual similarity, clustering, and retrieval tasks.

Consecutive embeddings are compared using similarity scores. When the similarity between sentences drops below a threshold, it is treated as a natural boundary, and a new chunk is created. This ensures that only semantically related sentences are grouped together. Additionally, a sliding window with overlap is applied so that a few sentences from the end of one chunk are repeated at the beginning of the next. This prevents loss of context at boundaries and improves the model's ability to maintain continuity. A maximum token length constraint (e.g., 300–800 tokens) is also enforced to keep chunks within the processing limits of large language models.

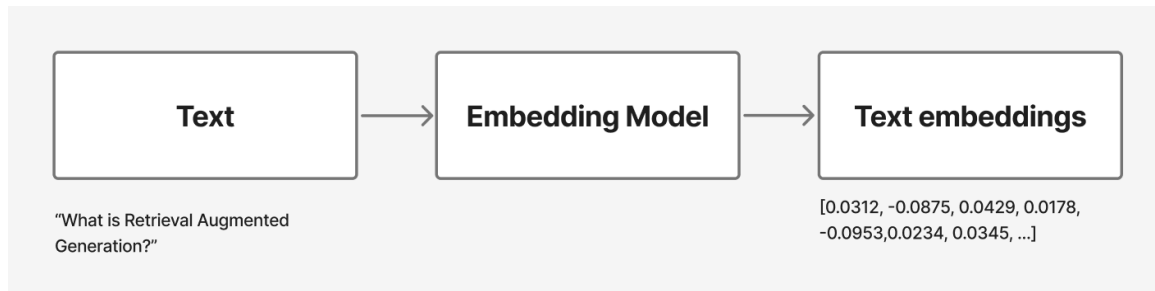
Embedding

After documents are divided into chunks, each chunk must be converted into a format that allows the retrieval system to efficiently compare their meanings. This is accomplished through embeddings. An embedding is a numerical representation of text in a high-dimensional space, where the distance between two vectors reflects their semantic similarity.

In this study, the Sentence Transformer MiniLM v6 was used to generate embedding. It is a lightweight yet powerful transformer-based model that achieves performance comparable to larger models (such as BERT or RoBERTa) while requiring significantly fewer computational resources. The “v6” variant is a fine-tuned version of MiniLM, optimized specifically for semantic textual similarity, clustering, and retrieval tasks. The embedding steps are:

1. **Input Preparation** – Each chunk of text, typically consisting of several sentences, is passed into the embedding model.
2. **Transformation** – The model converts the input into a fixed-length vector of floating-point values (e.g., 384 or 768 dimensions). This vector numerically encodes semantic properties of the text.
3. **Semantic Space Mapping** – The generated vectors are situated within a high-dimensional semantic space. In this space, chunks that discuss related concepts

(e.g., “*plants produce food*” and “*photosynthesis*”) are positioned close to each other, even if they use different wording



Vector Store

Embedded text chunks are stored in a vector database, and for this system we use FAISS (Facebook AI Similarity Search). FAISS is an open-source library developed by Facebook AI Research that is optimized for efficient similarity search and clustering of dense vectors. It is particularly well-suited for large-scale information retrieval tasks, as it can handle millions of embeddings while supporting both exact nearest neighbor search and approximate methods for faster lookups.

In this study, FAISS serves as the core vector index, enabling the system to quickly retrieve the most semantically relevant chunks given a query embedding. Its ability to perform high-dimensional similarity search (using metrics such as cosine similarity and L2 distance) ensures that the retrieval process is both accurate and efficient. T

Retrieval Module

The Retrieval Module is necessary for retrieving the relevant information for a user's query. The system incorporates both sparse and dense retrieval methods, which are combined into a hybrid retrieval approach to maximize accuracy and coverage. Once candidate results are collected, the system applies reranking to prioritize passages according to their true relevance to the query. To further enhance precision, thresholding is used to filter out weakly related results, ensuring that only high-quality and

contextually appropriate information is retained. Finally, the top-ranked passages are packaged as context and provided to the language model, enabling it to generate factually grounded and coherent answers.

Sparse Retrieval

Sparse Retrieval is a traditional method in information retrieval where both documents and queries are represented as sparse vectors, meaning most of the entries are zero since only the words that occur in the text receive non-zero values. Sparse Retrieval works by matching exact terms between the query and documents, which is made efficient through an inverted index. An inverted index is a data structure that maps each word in the vocabulary to the list of documents that contain it, allowing the system to quickly find all candidate documents for a query without scanning the entire collection. Once the relevant documents are retrieved, they are scored to determine how well they match the query. An issue with keyword matching is that simply counting word occurrences is not sufficient, since some words appear very frequently (like “the” or “is”) and are less informative, while rarer words provide stronger signals of relevance.

To handle this, the ranking function BM25 is used. BM25 is a Bag-of-Words model, where the presence or frequency of terms is used for matching this improves retrieval by balancing three key factors: the frequency of query terms within a document (term frequency), how rare or informative those terms are across the entire collection (inverse document frequency), and document length normalization so that longer documents do not automatically receive higher scores. In this way, sparse retrieval offers an efficient and interpretable approach to search, relying on keyword overlap enhanced with statistical weighting. We utilized BM25 Ranking for our Sparse Retrieval to make keywords comparison with the query

BM25 can be expressed as

$$BM25(D, Q) = \sum_{t \in Q} IDF(t) \cdot \frac{f(t, D) \cdot (k_1 + 1)}{f(t, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{avgdl}\right)}$$

Where:

- **d** = a document
- **q** = the query
- **t** = a term in the query
- **f(t,d)** = frequency of term *t* in document *d*
- **|d|** = length of document *d* (in words)
- **avgdl** = average document length in the collection
- **k, b** = tuning parameters (commonly $k \approx 1.2-2.0$, $b \approx 0.75$)

The BM25 IDF (t) formula uses a log function that reduces the impact of very rare terms and stabilizes scoring. The BM25 IDF is calculated as:

$$IDF(t) = \ln \left(\frac{N - n_t + 0.5}{n_t + 0.5} + 1 \right)$$

Where:

- t = a term
- N = total number of documents in the collection
- $n(t)$ = number of documents containing term t

Dense Retrieval

Dense vector search, also known as semantic retrieval, has emerged as a foundational technique in modern information retrieval systems. It represents queries and documents as low-dimensional, dense vectors using deep learning models such as BERT or other transformer-based encoders [42]. Unlike sparse retrieval like the BM25 function, which relies on exact word matches, dense retrieval uses semantic similarity, where words, phrases are mapped into a continuous vector space using neural networks this are then retrieved by using Retrieval is then performed by finding documents whose embeddings are close to the query embedding either by using dot product or cosine similarity. The relevant documents are compared in their similarity in a vector space, so exact matches are not relevant in the type of retrieval.

We used Cosine similarity for determining relevant documents within the Dense vector space. Cosine similarity is a metric that measures how similar two vectors are by looking at the angle between them, not their length. By focusing on vector orientation rather than magnitude it is effective in identifying semantic similarity in high-dimensional data, where document length or vector scale should not distort similarity.

In Cosine Similarity For two vectors, A and B:

$$\text{CosineSimilarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}.$$

Where:

- $A \cdot B$ is the dot product of the vectors,
- $\|A\|$ and $\|B\|$ are the Euclidean norms (magnitudes) of the vectors.

The resulting value ranges from -1 to 1 , where:

- 1 indicates that the vectors are perfectly aligned (high similarity),
- 0 indicates that the vectors are orthogonal or unrelated,
- -1 indicates that the vectors point in completely opposite directions.

Hybrid Retrieval

Hybrid Retrieval is an approach that combines both sparse retrieval methods and dense retrieval methods, to take advantage of their complementary strengths. Sparse retrieval excels at finding documents with exact keyword matches, while dense retrieval captures semantic meaning even when different words are used. In hybrid retrieval, we used score fusion, where the relevance scores from both sparse and dense systems are combined to produce a final ranking. For example, a document may receive a BM25 score based on keyword overlap and a cosine similarity score based on embedding similarity, and these scores are merged (through weighted addition, normalization, or other fusion techniques). Empirical studies demonstrate that hybrid retrieval consistently outperforms either method alone across a range of benchmarks [46, 47]. to determine the overall relevance. This way, hybrid retrieval ensures that documents which match the query both literally and semantically are ranked highest, improving accuracy and robustness compared to using either method alone.

The Fused score is expressed as:

$$FusedScore(d) = \alpha \cdot sdense(d) + (1 - \alpha) \cdot sBM25(d),$$

Where

Reranking

After retrieving candidate documents, we apply a reranking stage using a pretrained cross-encoder model named BAAI/bge-reranker-base base a cross-encoder reranker model developed by the Beijing Academy of Artificial Intelligence (BAAI). It belongs to the BGE (BAAI General Embedding) model family, designed for retrieval and reranking tasks in information retrieval (IR) and Retrieval-Augmented Generation (RAG) pipelines. Unlike bi-encoder used in dense retrievers, that transforms a query to find relevant documents into a vector space that independently encode queries and documents into embeddings for similarity comparison in vector space, a cross-encoder jointly processes each **query–document pair** and produces a **direct relevance score (logit)**. It then computes a **relevance score** that indicates how well the passage matches the query. This joint encoding allows the model to capture **richer semantic interactions** between the query and document text, which improves retrieval precision.

The output of the BAAI/bge-reranker-base cross encoder is unbounded. To make easy to interpret scores we applied a sigmoid transformation to map raw logits into a probabilistic like $[0,1]$ range:

$$s(d | q) = \sigma(\text{CE}(q, d)) = \frac{1}{1 + e^{-\text{CE}(q, d)}}$$

After the scores are normalized, we filter applied a thresholding technique > 0.3 to remove low-quality documents by applying a minimum score threshold . This removes weakly relevant candidates. By running a cross encoder to the query-passage pair, we ensured semantic relevance and precision. We then sent the top 10 documents from the Cross encoder reranked as context to the generation module.

Generation Module

Large Language Model

In this project, we used a pretrained large language model (LLM) called Mistral-Instruct to generate answers. Mistral is an open-source model family designed to be efficient and high-performing while remaining lighter than larger models such as GPT-3 or LLaMA-2. The “Instruct” variant has been fine-tuned on instruction-following datasets, which means it is better at understanding natural user prompts and producing context-aware, helpful responses rather than free-form text. By using Mistral-Instruct, we leverage a model that already carries strong general knowledge and language generation abilities without the need for training from scratch.

prompt Assembly

Prompt Assembly refers to the process of carefully constructing the input that is sent to the large language model before it generates a response. Since LLMs are highly sensitive to how instructions and context are phrased, assembling the prompt ensures that the model has the right information in the right format to produce accurate and useful answers. In our setup, prompt assembly typically involves combining several components: the user’s query, the retrieved documents (and possibly reranked passages), and a set of instructions or guidelines that tell the model how to use the information. For example, the assembled prompt might first present the user’s question, then include the most relevant pieces of text from the knowledge base, and finally add explicit instructions such as “Answer concisely based only on the provided context.” By structuring the prompt in this way, we reduce irrelevant or hallucinated responses and guide the LLM toward generating clear, grounded, and instruction-aligned outputs.

2.2.1 Windowing

As LLMs are known to skip important information in the middle, known as the middler curse. We utilized windowing technique wherein the most important retrieved context are placed in either at the top or bottom of the whole context block. This aims to alleviate the middle curse problem.

2.2.2 Prompt Engineering

We applied Prompt Engineering to design a template that guides the model to generate responses strictly based on the factual content provided in the context. The template ensures that if a user query is nonsensical, ambiguous, or unrelated to the retrieved

context, the model will refrain from answering and instead respond that it has no information about it. Figure 2 illustrates an example of this template. When submitted to the LLM, the prompt begins with clear instructions directing the model to only answer if the retrieved context is relevant to the question. If no relevant context is found, the model is instructed to reject the query gracefully. This method prevents the model from relying on its internal knowledge or fabricating answers, thereby reducing the risk of hallucination and improving factual accuracy.

```
prompt = f"""  
  
if context is empty or irrelevant say "I cannot answer that because I do not have information about it".  
  
Context:  
{context}  
  
Question:  
{query}  
  
Answer:  
"""
```

System Evaluation

The evaluation of the system follows a mixed-methods approach, combining quantitative performance metrics with qualitative user feedback to capture both the technical effectiveness and the practical usefulness of the platform. Because a Retrieval-Augmented Generation (RAG) system integrates two critical components, document retrieval and answer generation—it is important to assess these modules separately as well as in combination through end-to-end testing.

1. Retrieval Evaluation

The first step in evaluation is to measure how effectively the system retrieves relevant document chunks. This is critical because poor retrieval directly limits the quality of generated answers. For this purpose, we employ the following metrics:

- Precision@k – Measures the precision of the retriever by
- Recall@k – measures the proportion of all relevant chunks that are successfully retrieved within the top- k .

The method for this is by creating a gold standard dataset which is a compilation of query along with its relevant passages from the lecture notes we aim to use for evaluation

2. Generation Evaluation

Once relevant chunks are retrieved, the system must generate accurate and coherent answers. To evaluate the quality of generated responses, we rely on **automatic similarity-based metrics** that compare system outputs against reference answers (e.g., instructor-provided solutions or textbook explanations):

- **Faithfulness (Groundedness)** – assesses whether the generated answers remain consistent with the retrieved lecture notes.
- **Correctness (Accuracy)** – measures whether the response matches reference answers in meaning and content.
- **Relevance** – determines whether the generated text directly addresses the query.

Automatic Evaluation

- **ROUGE** – measures overlap between system responses and reference answers.
- **BLEU** – evaluates n-gram precision, often used for translation and summarization tasks.
- **BERTScore** – captures semantic similarity between system outputs and reference answers using embeddings.

This component-level evaluation provides a quantitative assessment of the system's ability to generate accurate and relevant answers grounded in the lecture material.

3. End-to-End Evaluation

We would also use End-to-End Evaluation by

- **Efficiency Metrics** – including average retrieval latency, response generation time, and token usage.
- **User Satisfaction** – gathered through feedback surveys focusing on ease of use, clarity of responses, and trust in the system's outputs.
- **Robustness Testing** – submitting ambiguous or adversarial queries to determine how well the system handles errors, irrelevant inputs, or attempts at prompt injection.

- **Task-based Evaluation** – participating students and developers attempt to answer study questions or complete tasks with the assistance of the system.

By combining these three layers of evaluation—retrieval quality, generation accuracy, and end-to-end user effectiveness—we ensure that the system is measured both quantitatively and qualitatively. This comprehensive evaluation framework highlights not only the technical performance of individual components but also the practical value of the system as a tool for information access and learning.

Ethical Consideration

Ethical considerations are utmost in conducting this research, which aims to improve students' educational experiences through technology. The following measures will be implemented to ensure the study is conducted responsibly and ethically:

1. Confidentiality: The privacy of participants will be strictly maintained.

Personal information, including any identifying data, will be anonymized to protect student identities. Data will be securely stored and only accessible by the research team. Anonymization ensures that results cannot be traced back to individual students, protecting their privacy, as emphasized in the IEEE Code of Ethics (Institute of Electrical and Electronics Engineers, 2020).

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2. Data Protection: The study will comply with the Data Privacy Act of 2012 (Republic Act No. 10173) of the Philippines, ensuring the secure and ethical handling of students' data. Collected data will be used solely for research

purposes, and participants will have the right to access or request the deletion of their data at any time.

APPENDICES

APPENDIX A. LETTERS AND DOCUMENTATIONS

APPENDIX B.

TRANSCRIPT OF

INTERVIEWS AND

PHOTO

DOCUMENTATION

APPENDIX C.

PANEL'S LIST OF RECOMMENDATIONS

REFERENCES